

Preclinical Molecular Docking & Pharmacokinetic Evaluation

Rigorous simulation conducted with AutoDock Vina v1.2.7, Meeko flexible torsion setup & RDKit cheminformatics

Investigational API / Ligand
Imatinib (C29H31N7O)

Biological Target Macromolecule
Crystal Structure of Human c-KIT Tyrosine Kinase in
Complex with STI-571 (Imatinib) (PDB: 1T46)

SECTION 1: EXECUTIVE QUANTITATIVE BINDING METRICS

Primary Simulation

Docking Mode: NATIVE REDOCKING / SELF-VALIDATION

Applicable (Crystallographic Self-Validation)

Target Receptor:
1T46 (Crystal Structure
of...)

Crystal Reference:
STI / STI-571

Docked Compound:
Imatinib

Self-Validation:
Pass (Research Grade)
(0.75 Å)

Imatinib was independently prepared and redocked into the receptor crystal structure after removal of the native co-crystallized ligand (STI / STI-571). Chemical identity confirmed by: InChIKey connectivity match (RCSB CCD): docked=KTUFNOKKBVMGRW-UHFFFAOYSA-N == native=KTUFNOKKBVMGRW-UHFFFAOYSA-N. This is a native crystallographic self-validation experiment.

Vina Docking Score
-12.39 kcal/mol
Empirical Score (ΔG estimate)

Affinity-Derived Kd-like
Estimate
0.83 nM
Derived / Model-based •
exp(score/RT)

Ligand Efficiency (LE)
0.335
|Score| / 37 heavy atoms

Intermolecular Contacts
3 H-Bonds
1 Salt • 2 π - π • 2 π -Cat • 0 Hal

Table 1: Standardized Biophysical & Docking Parameter Matrix

Explicit Scientific Grounding

Parameter	Value	Metric Type	Source / Methodology
Vina Docking Score	-12.39 kcal/mol	Predicted	AutoDock Vina 1.2.5 Empirical Scoring Function
Affinity-Derived Kd-like Estimate	0.83 nM	Derived / Model-based	Model conversion: $K_d = \exp(\text{score}/RT)$ at 298.15 K ($RT \approx 0.592$ kcal/mol). Not an experimental or rigorous thermodynamic K_d .
Ligand Efficiency (LE)	0.335 kcal/mol/HA	Calculated	Normalized Heavy Atom Affinity: $ \text{Vina Score} / 37$ Heavy Atoms
True Redocking RMSD	0.75 Å (Pass (Research Grade))	Validation	Symmetry-Aware Coordinate RMSD vs Co-Crystallized Ligand (≤ 2.0 Å Pass)
ChEMBL Wet-Lab Bioactivity	No deposited records	Experimental	Curated in vitro bioactivity records from the ChEMBL Database

* Note: The Affinity-Derived Kd-like estimate is mathematically derived from the AutoDock Vina score ($K_d = \exp(\text{score}/RT)$ at 298.15 K) and is not an experimentally measured or rigorously calculated thermodynamic K_d .

Protocol Self-Validation (Native Ligand Redocking):

RMSD = 0.75 Å • Pass (Research Grade)

SECTION 2: BIOPHYSICAL & SEARCH SPACE PARAMETERS

Algorithm: Vina Iterated Local Search

Target Receptor

PDB Identifier: **1T46**
Chains / Heavy Atoms: A • 2381 atoms
Grid Center [X, Y, Z]: **26.2, 26.1, 40.4**
Search Box Dimensions: 22.0 × 22.0 × 22.0 Å
Search Exhaustiveness: **32 (Deep Sampling)**

Investigational Ligand

Ligand Name: **Imatinib**
Molecular Formula: C29H31N7O
Molecular Weight: 493.62 g/mol
Heavy Atoms / Rot. Bonds: 37 heavy • 7 rotatable
Wildman-Crippen LogP: **4.59**

Biophysical Preparation & Catalytic Metalloenzyme Parameters

Biophysical Polar Hydrogen Geometry Engine (pH 7.4)

Catalytic Metals: **0 Retained**
(Zn/Mg/Mn/Fe/Ca)
Polar Hydrogens: **515 Added (AMBER FF14SB)**

Solvent / Buffer: **268 stripped / 10 buffer ions**
Forcefield / Charges: AMBER FF14SB / Kollman

Mode	Vina Score	VinarDO Score	Kd-like Estimate (Derived)	Ligand Eff.	Pose vs Rank 1 RMSD (l.b.)	Pose vs Rank 1 RMSD (u.b.)	Conformational Role
#1	-12.39 kcal/mol	-7.22 kcal/mol	830.7 pM	0.335	0.000 Å	0.000 Å	Rank 1 (Global Min)
#2	-11.53 kcal/mol	-6.56 kcal/mol	3.5 nM	0.312	1.007 Å	1.775 Å	Conformational Mode
#3	-10.32 kcal/mol	-6.86 kcal/mol	27.2 nM	0.279	3.394 Å	6.761 Å	Conformational Mode
#4	-10.13 kcal/mol	-5.10 kcal/mol	37.5 nM	0.274	5.680 Å	10.280 Å	Conformational Mode
#5	-9.91 kcal/mol	-5.22 kcal/mol	54.2 nM	0.268	2.417 Å	12.916 Å	Conformational Mode
#6	-9.67 kcal/mol	-4.34 kcal/mol	82.2 nM	0.261	1.819 Å	12.843 Å	Conformational Mode
#7	-9.41 kcal/mol	-5.10 kcal/mol	126.8 nM	0.254	2.361 Å	11.763 Å	Conformational Mode

**Methodology Distinction:* "Pose vs Rank 1 RMSD" quantifies internal conformational clustering of generated poses relative to the lowest-energy mode (#1). It is fundamentally distinct from crystallographic redocking RMSD, which measures coordinate deviation from a true wet-lab crystallographic reference structure.

SECTION 4: INTERMOLECULAR INTERACTION FINGERPRINT

PLIP-Standard Non-Covalent Binding Contacts

Directional Hydrogen Bonds (Cutoff ≤ 3.5 Å): 3 Detected

Receptor Residue	Receptor Atom	Ligand Atom	Distance (Å)	Bond Type
ASP 810:A	N	O	2.91 Å	Hydrogen Bond
THR 670:A	OG1	N	3.36 Å	Hydrogen Bond
CYS 673:A	N	N	2.92 Å	Hydrogen Bond

Electrostatic Networks & Salt Bridges (Cutoff ≤ 4.2 Å): 1 Detected

Ionized Residue	Pairing Classification	Ligand Atom	Distance (Å)	Criterion
GLU 640:A	Ligand Cation - Protein Anion	N	3.62 Å	Cutoff ≤ 4.2 Å

π-π Aromatic Stacking (Cutoff ≤ 5.5 Å): 2 Detected

Residue	Geometry	Distance	Angle (°)
TYR 672:A	Parallel π-π	5.18 Å	22.0°
PHE 811:A	T-shaped π-π	4.57 Å	71.9°

π-Cation Contacts (Cutoff ≤ 4.5 Å): 2 Detected

Residue	Classification	Ligand Atom	Distance
TYR 672:A	Ligand Cation - Receptor π-Ring	N	4.23 Å
LYS 623:A	Receptor Cation - Ligand π-Ring	Ring (6 atoms)	4.12 Å

Halogen Bonds (σ-Hole Cutoff ≤ 3.8 Å, θ ≥ 130°): 0 Detected

Receptor Residue	Receptor Atom	Ligand Atom	Distance (Å)	Angle (C-X...O/N)
No halogen bonds or polar fluorine contacts detected within distance (≤ 3.8 Å) and geometric criteria.				

Hydrophobic Contact Residues (Cutoff ≤ 4.0 Å): 16 Contacts

LEU 644:A (3.90 Å)	VAL 654:A (3.72 Å)	GLU 640:A (3.91 Å)	LEU 644:A (3.36 Å)	LYS 623:A (3.87 Å)
THR 670:A (3.31 Å)	LYS 623:A (3.70 Å)	VAL 668:A (3.50 Å)	THR 670:A (3.55 Å)	THR 670:A (3.56 Å)
GLU 640:A (3.77 Å)	LEU 644:A (3.34 Å)	VAL 668:A (3.81 Å)	THR 670:A (4.00 Å)	THR 670:A (3.98 Å)
ALA 621:A (3.93 Å)				

Lipinski Parameter	Value	Criterion	Status
Molecular Weight (MW)	493.62 Da	≤ 500 Da	Pass
Wildman-Crippen LogP	4.59	≤ 5.0	Pass
H-Bond Donors (HBD)	2	≤ 5	Pass
H-Bond Acceptors (HBA)	7	≤ 10	Pass

Pharmacokinetics & Bioavailability

TPSA (Topological Polar Surface):	86.28 Å² (≤ 140 Å²)
Rotatable Bonds Count:	7 (≤ 10)
Gastrointestinal (GI) Absorption:	High
Blood-Brain Barrier (BBB):	Non-Permeant
CYP450 Liability:	No Inhibitory Alerts

ACADEMIC ATTRIBUTION & SOFTWARE REPRODUCIBILITY

Calculation Timestamp: Fri, 25 Sep 2026 18:15:34 GMT

Docking Engine:
AutoDock Vina 1.2.5
(Scripps CCSB)

Cheminformatics:
RDKit v2026.03.6

Crystallography:
Gemmi v0.7.5

PDBQT Setup:
0.5.x
(MoleculePreparation /
PDBQTWriterLegacy)

Citation Notice: If referencing computational predictions generated by this platform in an academic dissertation or publication, cite AutoDock Vina (Trott & Olson, 2010), Vinardo (Quiroga & Villarreal, 2016), RDKit (Landrum et al.), and EMBL-EBI ChEMBL (Gaulton et al., 2017). Support inquiries: sharmaji.pharmatech.info@gmail.com.